SPACE PHYSICS
FOIL FLOATING
PROCESS

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1. INTRODUCTION

This procedure is designed to provide the user with basic steps for floating ultra-thin carbon foils. All carbon foil floating is to be done in a class 10,000 or better cleanroom with user wearing full cleanroom attire. Operator should put their initials and date for all operations as well as N/A where applicable. Operator should also document any variations from procedure in the 'notes' section provided at the end of every page. It may be necessary to vary from procedure slightly in order to achieve desired results.

2. MATERIALS AND TOOL CHECKLIST

• Carbon Foils	• Substrates
• Deionized Water (DI H2O)	• Isopropyl Alcohol (IPA)
• Specific Gravity Hydrometer	• Graduated Cylinder
• 4000ml Beaker	• Glass Mixing Rod
• Syringe	• Scalpel
• Small Petri Dishes (labelled)	• Bus Wire
• Thermometer	• Tweezers
• Ruler	• Hot plates
• Spudger	• Tex wipe (small conical qtips)
•	•

Room Temperature : ____________________

Humidity : _______________

Particle Counter reading : _______________

Operator :_____________ Date ________________

3. DI H20 – IPA MIXTURE

1. Use the 4000 ml beaker with the magnetic stirrer to mix DI H2O-IPA, (7 -8 % IPA) for 2-3 minutes (as seen in Fig.1) till the solution seems homogenous. The IPA concentration in the solution can depend on carbon foil thickness that is to be floated (IPA decreases water surface tension but also decreases floatation of carbon foil). The table below shows the component volumes for a 7% concentration, if using a different concentration, please note the values below.

Water Volume	2790 ml
IPA (99%) Volume	210 ml
Total Solution Volume	3000 ml
Final Mixture %	7%

2. Set the temperature on the stirrer hot plate to 60-80 deg C since we need to heat the solution.
3. Measure and confirm the concentration of DI H2O-IPA mixture using the specific gravity hydrometer (see Fig.1 middle panel).
4. Once mixed, transfer the solution to the glass tray to be used for floating the foil.
5. Turn the hot plate under the glass tray on and set the temperature to 60 degC initially. This temperature needs to be adjusted in order to maintain the temperature of the bath to be between 35-36 degC. Do not overheat DI H2O-IPA solution to more than 40 degrees Celsius, as this will create bubbles in water, and subsequently holes in the carbon foil. Ideal temperature is between 35-45 deg C. Verify and record the temperature using a thermometer.

FINAL TEPERATURE OF THE SOLUTION (deg C):______________________

OPERATOR:___________________________ DATE:____________________

Notes-

Fig.1 Beaker with magnetic stirrer (LEFT), graduated cylinder with specific gravity hydrometer (CENTER), thermometer measuring the temperature of the solution bath (RIGHT).

4. SUBSTRATE SURFACE PREPARATION

The objective of substrate surface preparation is to clean substrate surface and promote the adhesion of carbon foil to substrate surface, sufficient for carbon foil to survive space flight environmental testing. The substrate surface needs to be in a hydrophilic state (water readily wets the surface). If the substrate surface is hydrophobic, water will tend to bead-up on substrate surface. Carbon foils tend to avoid adhering to a hydrophobic surface. A simple cleaning of the substrate surface with isopropanol (99%) (no ultrasonic bath) is all that is necessary to achieve proper adhesion of carbon foil. The substrate commonly used in this experiment is a Nickel (Ni) grid in a Ni washer (as shown in Fig 2). Each grid can be scribed with a serial number (usually the date followed by an alphabet) in the case of floating multiple foils together. This step needs to be done prior to the cleaning step.

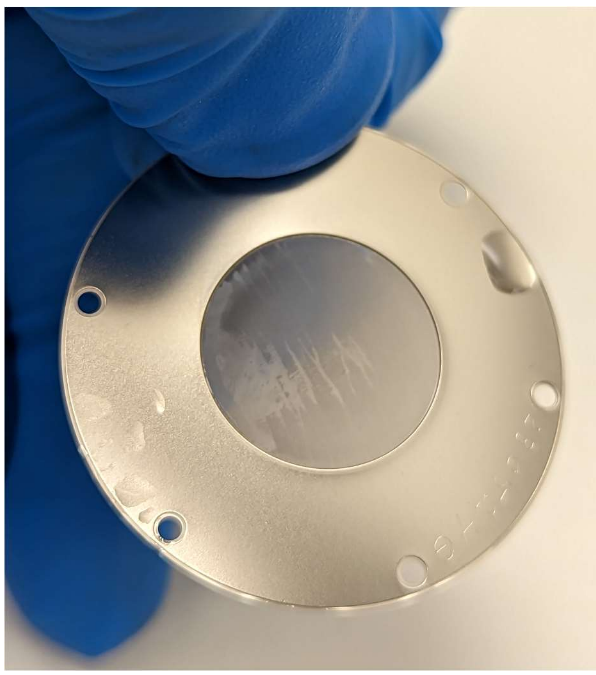

Fig.2 Nickel substrate post-cleaning ready for carbon foil floating. The scribed serial number can be seen at bottom right of the Ni grid.
Please note down the specifications below –

SUBSTRATE MANUFACTURER	Thin Metal Parts (TMP)
SUBSTRATE TYPE (lines per inch)	333 lpi
SUBSTRATE NUMBER	

If floating multiple foils, use the table in Section 8 to fill out more details.

5. CARBON FOIL PREPARATION

We obtain the Carbon films from Arizona Carbon Foil (ACF) company. The Arc evaporated carbon foils come on glass substrates, hand polished with parting (or release) agent to permit removal by floating. ACF has two standard sizes of glass substrates, 25mm x 70mm and 50mm x 70mm, provided in individual protective containers by the company. The areal density of the foils, also known as surface density, specific density, or mass per unit area, is in units of micrograms per square centimeter. Foils thickness is measured optically and/or by weighing in order to provide maximize accuracy by the company. The Carbon film thickness ranges from 0.5 to 3µg/sq.cm. For purposes of SWAPI, we mainly use the thickness of 1µg/sq.cm.

A picture of the box containing the glass slides with Carbon film is shown in Fig 3. Each glass slide itself has a serial number and the thickness marked on it (Fig 3).

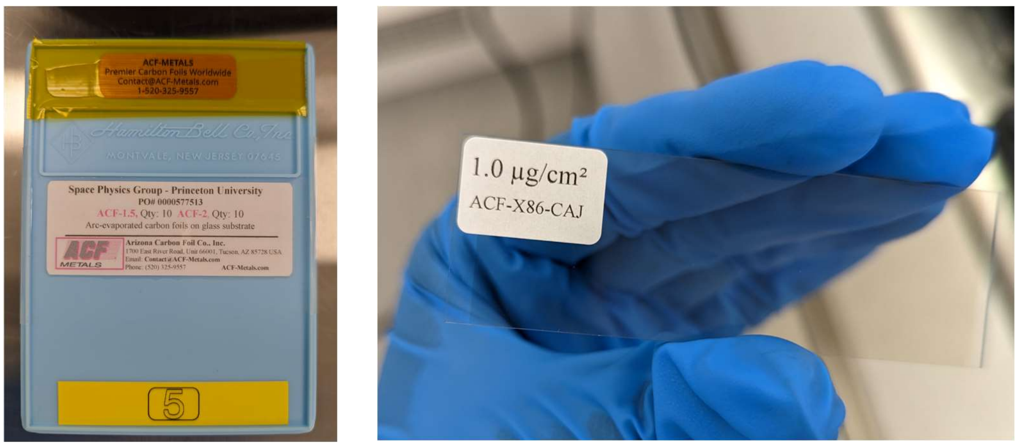

Fig.3 Box containing the Carbon Foils from Arizona Carbon Foil company (LEFT), the said box number is 5 as seen at the bottom. The actual glass slide has a fine layer of deposited Carbon and a sticker stating the thickness of the film and a serial number (RIGHT).

GLASS SLIDE BOX NUMBER	
GLASS SLIDE DIMENSION	
GLASS SLIDE QC NUMBER	
CARBON FILM THICKNESS	

If floating multiple foils, use the table in Section 8 to fill out more details.

PROCESS :

1. Carefully scribe the edges of the carbon foil on the glass slide using a sharp scalpel (see Fig. 4). Make sure you do not touch the extremely fragile Carbon film. This is to prevent the foil edges from being stuck to the slide during the float. Also, scribe through the center if you plan on floating two foils from the same glass slide.

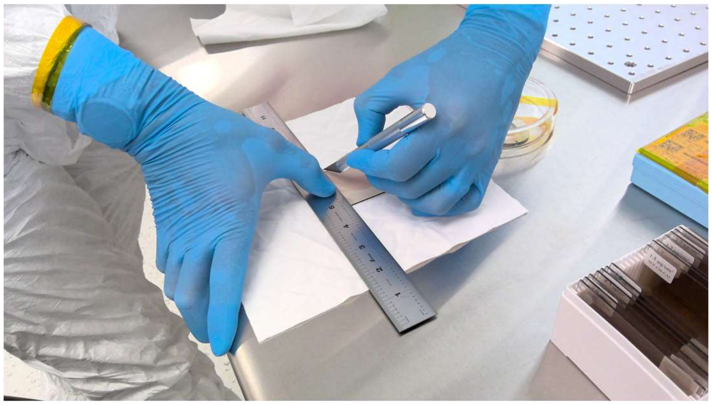

Fig.4 Scribing the Carbon Foil

2. Clean the edges of the glass slide using a small Q-tip. This is done to remove any residual carbon (or crud) on the sides that may prevent a smooth release of the carbon foil.
3. Use a wet alpha wipe in a large petri dish to act as a humidifier where the glass slide can be stored after scoring and before installing onto the stage. This has been observed to help in a smoother release of the carbon foil.

6. FOIL FLOATING STATION

The carbon foil floating station consists of 2 stage assemblies. Each stage assembly incorporates 5 ranges of motion (X-axis, Y-axis, Z-axis, Rotational-axis, and Feed-axis). For the purpose of this procedure the LEFT stage will be the substrate (Ni grid) stage and the RIGHT stage will be the carbon foil glass slide stage (this can be swapped based on user preference).

The X, and Z-axis stages are generally used for rough positioning of carbon foil slide and substrate and may be manipulated as necessary. X-axis is utilized for left to right positioning of carbon foil slide and substrate. Z-axis is utilized for up and down positioning of carbon foil slide and substrate. Y-axis is utilized for lateral positioning of carbon foil slide and substrate.

The Feed axis on both stages incorporate an adjustable extension rod which allows user the option of extending or retracting the carbon foil slide and the substrate in order to achieve optimum positioning. The Feed axis stage is used for mounting the carbon foil and substrate and for feeding the carbon foil and substrate into the DI H2O-IPA solution.

Rotational axis is utilized for rotating carbon foil slide and substrate in and out of the DI H2O-IPA solution, and for angular adjustment of carbon foil slide and substrate. This is to ensure that both the carbon foil slide and the substrate have are not at an angle with respect to the solution surface.

The floating bowl sits on top of the hot plate and is used to control the temperature of the DI H2O-IPA mixture as described in Section 3. Fig.5 shows the foil floating set-up with all its different components.

7. FOIL FLOATING PROCEDURE

1. Mount the nickel grid first, so there the carbon glass slide can stay in the humidifier longer.
2. Utilizing the Substrate motion stages, position the substrate into the DI H2O-IPA solution to where entire effective area of substrate is submerged in DI H2O- IPA solution. (The Substrate Z-axis stage may also have to be used).
3. Mount the carbon glass slide onto the motion stage holder next. Utilizing a combination of Carbon foil Feed stage, and Carbon foil X-axis stage, position the glass slide to where it is as close to the substrate as possible.
4. Now **slowly** feed the slide's leading edge into the DI H2O- IPA solution.
5. Use the edge of the carbon film slide to observe the solution meniscus and how it releases from the slide in order to ensure that the glass slide is parallel to the solution. We don't want the slide to enter the solution at an angle (Fig 6).

CAUTION: DO NOT FEED CARBON TOO FAST, AS THIS MAY CAUSE WRINKLES OR FOLDING OF CARBON FOIL.

6. Note the stage positions for future reference in the table below –

Ni Grid Rotational Stage	73.7 deg
Glass Slide Rotational Stage	209 deg

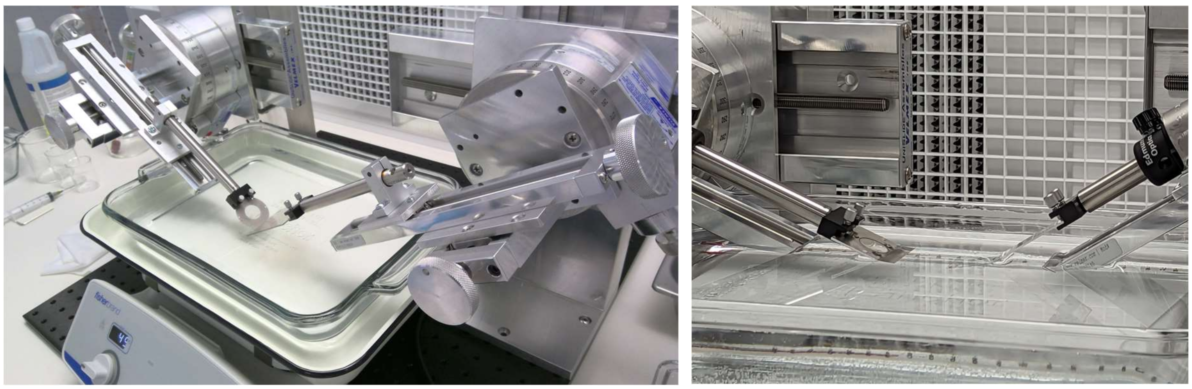

Fig. 6 Foil Floating in progress with the Ni substrate and glass slide being moved into position (LEFT). Zoomed-in view of the glass slide just touching the solution surface (RIGHT).

7. Very **slowly** continue to feed the slide into DI H2O- IPA solution and observe as the carbon foil starts to release from slide. until it is fully released (Fig. 7).
8. Use **Section 8** to make note of different times regarding the touching and releasing of the Carbon film.
9. If required, use Substrate Y stage to adjust the substrate laterally under the foil, selecting the best portion of the carbon foil. Make sure there are no obvious holes or cracks in the selected portion of the foil.
10. Once optimum placement is achieved **slowly** rotate the Substrate Rotational stage up and out of DI H2O-IPA solution (Fig 7). Remove the substrate out of the holder.

Fig. 7 Carbon Foil starting to release (LEFT). Ni substrate being completely covered with the Carbon foil, right after being pulled out of the solution (RIGHT).

OPERATOR:______________________ DATE:_________________

NOTES:

8. EXPERIMENT LOG

Record the following entries during the experiment –

Ni Grid #	Carbon Slide #	Start Time of Carbon film touching the solution	Start Time of Release of Carbon film	Start time of Contact between Ni grid & Carbon film	Time of removing the Ni grid completely out of the solution

9. DRYING PROCESS

1. Hang the carbon foil-substrate assembly on oven's wire shelf using bus wire. Substrates should not be set on the bottom floor of the oven as the floor is at a much higher temperature than the rest of the oven.

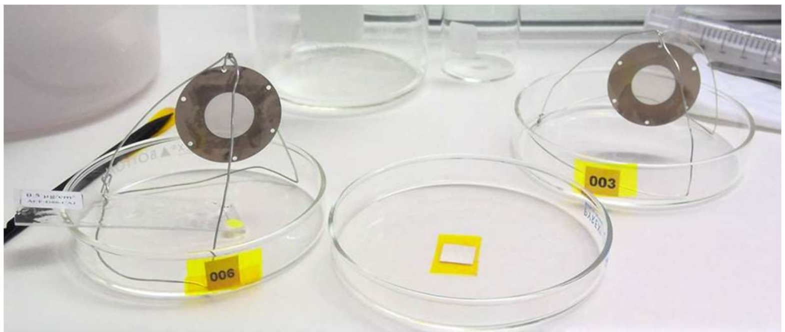

Fig 8. Floated foils put on bus wire frames (TOP), ready to be put in the oven.

2. Once all the floated carbon foils are in the oven, set the temperature to 50 degrees Celsius and turn it on. Allow oven to ramp up to the set temperature. This prevents any thermal shock to the carbon foils. Allow the assembly to dry for a minimum of 6 hours at 50 degrees Celsius.
3. To remove carbon foil-substrate assembly, turn the oven off and allow the temperature to ramp down to room temperature in order to prevent thermal shock.

TIME (IN): ________________ OVEN TEMPERATURE:________________

TIME (OUT): ________________ ROOM TEMPERATURE:________________

OPERATOR: ______________________ DATE:________________

NOTES:

10. CLEANING DI H20-IPA SOLUTION

It is important to maintain clean DI H2O- IPA solution throughout the floating process. Maintaining clean a DI H2O- IPA solution will help eliminate loose particles of carbon migrating to the substrate and causing double layers of carbon on substrate.

Using an alpha-wipe large loose carbon debris can be removed from the solution.

Smaller particles of carbon can be removed by suctioning carbon debris off of DI H2O-IPA solution using a syringe (shown in Fig.9)

Fig.9 Using syringe to remove Carbon film debris from the DI H2O-IPA solution

For flight applications it is recommended DI H2O-IPA solution be partially or completely replaced between every carbon-substrate assembly floated.

For non-flight applications, DI H2O-IPA solution can be partially or completely replaced after every 3 or 4 floats. User will need to monitor DI H2O- IPA solution cleanliness to best determine when to change DI H2O- IPA solution. Before floating another carbon foil, verify DI H2O-IPA solution after carbon debris cleaning process.